Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

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Abstract

The field of *AI Scientist* systems—agents that generate hypotheses, execute experiments, and synthesize knowledge—lacks a shared formal vocabulary. We propose Servo (S, G, E, V, M, π), a six-component taxonomy comprising the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy (π), grounded in Partially Observable Markov Decision Process (POMDP) and Bayesian Experimental Design (BED) theory. The framework's core contribution is three diagnostic separations that current systems conflate: the observation likelihood from the validator as a reward channel; calibrated gating (V_{gating}) from mere layer presence (V_{present}); and episodic from semantic memory. We formalize these as three propositions under tractable, identifying assumptions—the loop closes independently of the validator; a miscoupled validator can drive convergence to a misspecified target; and a novelty component absent from every automated channel is non-identifiable (Propositions 1–3)—which no surveyed system satisfies, so they clarify the formal structure of the distinctions rather than certify their empirical prevalence. We apply it to six core systems and seven domains and identify three open problems with falsification criteria: the lack of validated automated novelty gates, the BED–practice gap for search policies, and the experiment fidelity gap (no principled criterion for escalating from computational proxy to physical validation). The cross-system survey is offered as provisional structured annotation; the organizing hypothesis that gating calibration co-occurs with trustworthy closure is direction-consistent under some outcome-recoding assumptions but not adjudicated, and should not be read as an empirical finding.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Systems range from tool-augmented synthesis agents (Coscientist [Boiko et al., 2023]) through closed-loop manuscript-writing pipelines (The AI Scientist [Lu et al., 2024, 2026]) to natural-language mathematics agents (Aletheia [Feng et al., 2026]), each introducing bespoke terminology that makes systematic cross-system comparison difficult. Without a shared decomposition, questions that bear directly on deployment remain unanswered: *What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?*

We propose Servo[1] (S, G, E, V, M, π), a formal framework grounding AI Scientist systems in POMDP and Bayesian Experimental Design theory (Section 4). We apply it to analyze six core

[1] We name the framework after the *servomechanism*: as a servo loop converges on a target by acting on a feedback (error) signal, an AI Scientist closes the discovery loop only when its validator V supplies a reliable feedback signal—the framework's central concern.

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systems, an artifact infrastructure proposal, and instantiations across seven scientific fields (Sections 5, 6); and identify three open problems with falsification criteria (Section 7). The analysis reveals a recurring diagnostic distinction: systems that mechanically close the loop differ sharply in whether their acceptance gate, novelty judgment, and memory structure can support trustworthy discovery claims; we develop the supporting theory in Appendix A. The contribution is thus a diagnostic vocabulary that keeps mechanical closure, validator calibration, novelty judgment, and memory separable rather than collapsing them into a single notion of autonomy.

Scope. We include systems whose primary goal is automated scientific *discovery* through hypothesis-experiment-validation cycles, and exclude specialized predictors without a generation-and-search loop (e.g., AlphaFold [Jumper et al., 2021]). Within that scope we analyze widely-cited representative systems rather than an exhaustive enumeration; this convenience sampling introduces selection bias that we flag in Section 8.

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as *partially observable Markov decision processes*: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O}, b_0, \gamma)$ with action space $\mathcal{A}$ (hypothesis generation and experiment execution) and reward R (scientific value of discovery); the belief over s is updated by Bayes' rule, and π acts on that belief rather than the inaccessible state s. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). We therefore use the POMDP as an organizing formalism, not a fully specified model of open-ended science: the results of Appendix A hold for the tractable subproblem (Assumption 1), which no surveyed system satisfies in full, so they apply to an idealized subproblem rather than directly to any system analyzed here. Throughout, three distinct senses of "loop closure" should be kept separate: (i) posterior concentration on $\theta^\star$ in the tractable subproblem (Propositions 1–3); (ii) *computational* closure, where feedback from V flows back into G; and (iii) *trustworthy* closure, where the acceptance gate reliably tracks truth. The propositions establish (i); the survey concerns (iii); systems may achieve (ii) without (i) or (iii). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides a normative criterion for the information objective of the search policy π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \mathrm{EIG}(d) = \arg\max_{d} \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Domain-specific systems already use active-learning approximations—the Robot Scientist's experiment selection [King et al., 2004] and GNoME's DFT-in-the-loop active learning [Merchant et al., 2023]—but recent open-ended LLM-based systems maintain no explicit posterior or stated EIG objective; that narrower gap is the one we identify in Section 7.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop; knowledge-graph-driven ideation systems [Gu and Krenn, 2024, Lee et al., 2025] further show that structuring semantic memory as a relational corpus can help surface and rank higher-interest research ideas, but equally without closing the loop.

Autonomous-science frameworks. Closed-loop autonomous discovery predates the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation, and statistical validation in yeast functional genomics, instantiating all six Servo components—including an active-learning policy [King et al., 2004]—in physical wet-lab form more than a decade before

LLM-based systems. More recent unified frameworks include NovelSeek [Zhang et al., 2025], a closed-loop multi-agent system spanning idea generation to verification across twelve scientific tasks, and Co-Scientist [Gottweis et al., 2026], a multi-agent hypothesis-generation system with experimental biomedical validation. We therefore position Servo not as the first unified framework but as a formal vocabulary making validator completeness, the computational-to-physical boundary, and the POMDP/BED relationship explicit for the LLM-agent subfield. Its diagnostic value is illustrated, not independently tested, at two points. First, standard POMDP classification and NovelSeek's multi-agent framework distinguish loop-closed from loop-open but do not separate V_{present} (a calibrated layer exists) from V_{gating} (the decisive gate is calibrated); both code AI Scientist (Nature 2026) as closed-loop, but Servo's V-layer decomposition locates the failure—the gating layer is biased, not absent. Second, existing POMDP treatments collapse loop closure into a binary, while Servo's three-way separation (posterior concentration, computational closure, trustworthy closure) identifies systems such as Agent Laboratory that close a computational loop without trustworthy gating.

Decision-theoretic foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical, with a long history of Bayesian-optimal experiment selection in automated science—most directly the Robot Scientist's active-learning policy [King et al., 2004]; we connect POMDP (structure) to BED (normative policy) for the LLM-agent setting rather than claiming priority for the idea. The reward-specification and specification-gaming literatures study unreliable reward signals; Servo extends this lens by stratifying V into reliability levels tied to specific closed-loop failure modes.

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding (CPC-MS) as a social model of science, recasting scientific activity as decentralized Bayesian inference across a community. CPC-MS and Servo are complementary: CPC-MS provides a *collective* and *social* model in which peer review is modeled as a Metropolis-Hastings acceptance (consensus) process; Servo provides the *single-system* and *modular* decomposition that must be instantiated at each node. The three open problems we identify constitute core implementation challenges for such a collective, viewed through Servo. Liu et al. [2026] address the artifact layer — output format and verification protocol — rather than the discovery loop itself; we position ARA as a V infrastructure contribution within Servo (the Seal automates V_{syntax}, V_{semantic}, and $V_{\text{empirical}}$), discussed in Sections 5.1 and 7.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S**: Search Space** The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \to h$**: Hypothesis Generator** A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \to o$**: Experiment Executor** A function that executes hypothesis h under protocol P to produce observation o. The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \to r$**: Validator** A function that assigns a reward signal r to observation o. We distinguish four layers of V:

$$\begin{aligned}
V_{\text{syntax}} &: o \to \{\text{pass}, \text{fail}\} && \text{(deterministic, fast)}\\
V_{\text{semantic}} &: o \to r \in [0,1] && \text{(LLM-based, stochastic)}\\
V_{\text{empirical}} &: o \to r && \text{(reproduction-based)}\\
V_{\text{human}} &: o \to \{\text{novel}, \text{significant}\} && \text{(human-delegated)}
\end{aligned}$$

M**: Memory** Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π.

$\pi : b \times M \to h$**: Search Policy** The decision rule for selecting the next hypothesis $h \in S$. The belief state b is updated after each experiment. In the BED interpretation, $\pi^* = \arg\max_d \mathrm{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned} \pi(b, M) \;\to\; G(M_s, S) \;\to\; E(h, P) \;\to\; V(o) \\ \to\; M \mathrel{+}= (h, o, r) \;\to\; b\,\text{update} \;\to\; \text{repeat} \end{aligned} \tag{3}$$

We further introduce $H \in [0, 1]$ (distinct from the entropy function $H(\theta)$ in Eq. 1) as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π.

V-completeness as a design dimension. The layers form a partial order under reliability: V_{syntax} deterministic; V_{semantic} stochastic and bias-prone; $V_{\text{empirical}}$ from well-calibrated (DFT) to poorly calibrated (GNN surrogates); V_{human} irreplaceable for novelty but rate-limiting; and V_{formal} (proof kernels, type-checkers) the mechanically-decidable ideal limit, used for mathematics (formal) in Table 2. We define V *completeness* as the highest layer both accessible and reliably calibrated; accessibility without calibration does not count (Agent Laboratory's biased reviewer yields only imperfect completeness). We treat calibration of the gating layer as the operative design variable—the holistic ordinal being exploratory (Appendix B)—because it caps what π can learn and what G can be rewarded for: without a reliable V, no policy can distinguish progress from regression. What matters is not that a calibrated layer is *present* but that the layer which *gates acceptance* is calibrated: a calibrated check can be neutralized when a biased semantic or social layer decides acceptance (§8). The cross-domain pattern (Section 6) appears to track the calibration of the gating validator more than G or π sophistication; we develop the relevant formal results next.

The validator as a reward channel. Treating V as distinct from the observation likelihood, we prove three results in a tractable, identifying subproblem (Appendix A): under identifiability and divergent accumulated information the loop closes (Proposition 1); a validator miscoupled into the belief update can drive convergence to a misspecified target (Proposition 2); and a novelty component invisible to every automated channel is non-identifiable (Proposition 3). A complete validator is thus *necessary* for trustworthy autonomous closure—i.e., for truth-tracking reward assignment—but, by Proposition 1, not sufficient without enough accumulated information; the surveyed systems illustrate these results rather than prove them.

5 Analysis of Core AI Scientist Systems

We apply Servo to the core systems in Table 1 and one artifact infrastructure proposal, synthesizing the key dimensions; in this small sample, the systems that reach closed-loop operation are also those with more complete validators, although they also advance G, E, and π (we revisit this interaction in Section 8).

Across these four LLM-based end-to-end systems (Coscientist, the two AI Scientist systems, and Agent Laboratory), those that close the loop are also those that upgrade or add validators; G or π advances are not absent—several also upgrade ideation, tool use, and search—but validator reliability is the constraint gating whether such advances can be exploited in a closed loop. Coscientist has only constrained task-level feedback (measured outcomes inform later choices [Boiko et al., 2023]); its loop is *partial*, with novelty/significance validation human-mediated. The AI Scientist (Sakana) adds V_s via a 5-ensemble GPT-4o reviewer (balanced accuracy 0.65 vs. human 0.66 [Lu et al., 2024]), enabling the first *structurally* closed loop on a biased, uncalibrated V. The AI Scientist (Nature)[2] stacks V_e (replication nodes with mean ± s.d.), V_s (automated reviewer, balanced accuracy 0.69 [Lu et al., 2026]), and actual peer review (V_h via ICLR workshop), giving the most layers in the class—though

[2]The AI Scientist's (2024) and (2026) variants are its template-based and agentic-tree-search ("AI Scientist-v2" [Yamada et al., 2025]) modes, the latter its peer-reviewed Nature version [Lu et al., 2026].

	Robot Sci.	Coscientist	AI Sci. (2024)	AI Sci. (2026)	AgentLab	NovelSeek
Novelty measure	Stat. test	None	LLM self-score	3-layer	Human-deleg.	Perf. gain
Loop closed?	Yes (wet-lab)	Partial	Yes (greedy)	Yes (tree)	Partial	Yes (comp.)
π type	Active learn.	ReAct	Greedy	Tree search	Human/seq.	Feedback
V completeness	V_e	$V_h + V_e$	V_s (biased)	$V_e + V_s + V_h$	V_s (biased)	V_e
M structure	$M_e + M_s$	Ephemeral	M_e	$M_e + M_s$	M_e	M_e
H (approx.)	Low	G (≈1)	Low (≈0)	Low (≈0)	G+π (≈0.8)	Low

Table 1: Comparative Servo analysis of core AI Scientist systems, including the pre-LLM Robot Scientist [Sparkes et al., 2010] and the recent unified framework NovelSeek [Zhang et al., 2025]. V_s=semantic, V_e=empirical, V_h=human. For AI Scientist (Nature 2026), V_s is present but non-gating; the decisive acceptance gate is V_h (ICLR peer review), consistent with `V_gating=0` in the released coding (`analysis/systems.csv`).

its automated *internal* acceptance gate (the biased V_s reviewer) stays uncalibrated (§8). Agent Laboratory delegates V to V_s (NeurIPS-guideline LLM reviewer) but documents $+2.3$-point systematic over-estimation relative to the human PhD-student reviewers of the same AI-generated papers—adding a V layer does not guarantee completeness if the layer is unreliable [Schmidgall et al., 2025]. Four end-to-end systems, however, cannot establish general impossibility.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology (`/logic`, `/src`, `/trace`, `/evidence`); the ARA Seal automates V_{syntax}, V_{semantic} (Rigor Auditor, 82.6% detection), and $V_{\text{empirical}}$ (sandboxed reproduction), reserving V_{human} for significance and novelty [Liu et al., 2026]. Motivating figures: only 45.4% of reproduction requirements are fully specified and 90.2% of extension cost is failed exploration [Liu et al., 2026]. The `/trace` layer externalizes M_e/M_s, but preserved traces both accelerated and anchored downstream agents—richer M_e does not monotonically improve π.

6 Domain Analysis

Table 2 is an illustrative domain map: it uses the Servo vocabulary to expose how validation bottlenecks differ across seven scientific fields, the bottleneck following from each field's V type (Propositions 1–3; per-domain analysis with sources in Appendix C). Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024, Kamber, 2026], materials [Merchant et al., 2023], biology [O'Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023a], and CS/algorithms [Real et al., 2020, Surina et al., 2025, Jaber and Jaber, 2026].

The pattern is largely shaped by V type. Formal mathematics has a binary V_{formal} that makes RL stable and the loop fully closed; natural-language proof reintroduces V_{human} (Aletheia: 6.5% audited-correct). Chemistry and biology close the computational loop but leave the physical loop open at the measurement boundary, where characterization stays human-delegated. Materials science is the surveyed domain whose policy most resembles active experimental design within a tractable computational surrogate: GNoME uses model uncertainty and DFT relaxations to close a computational loop for phase-stability prediction. But DFT approximates physical energies rather than being a fully calibrated oracle, and most predicted compounds still require synthesis and characterization [Cheetham and Seshadri, 2024]; the physical loop remains only partly closed. Social science suffers circular validity when G and E share a model, and CS/algorithms close the loop via reliable V_{emp} but lack interpretability.

7 Open Problems

Our analysis identifies three open problems that recur across the surveyed systems and illustrative domains. For each we state a *proposed* falsification test—a concrete design whose outcome would refute the claim—specifying what evidence would settle the question; none are executed here.

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Closed (low quality)	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High (comp.)	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial/ablation	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Illustrative Servo domain mapping across seven fields (per-domain coding in `domain_systems.csv`). V reliability is an ordinal calibration label, not measured agreement with ground truth. Domain codings are single-coder interpretations of the cited primary sources; no inter-coder κ statistics are available for this table.

7.1 The Lack of Validated Automated Novelty Gates

No automated validator among the surveyed systems reliably evaluates novelty—every surveyed system, from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation), converges on this. Two failure modes are documented: LLM self-evaluation bias (systematic over-estimation; hallucinated results) and pretraining contamination (systems may "discover" results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal's Rigor Auditor [Liu et al., 2026] achieves 82.6% average detection across five structural violation types but only 22% on orphaned-experiment violations; it cannot cross the novelty boundary. Concretely: Agent Laboratory's LLM reviewer over-estimated paper quality by $+2.3$ points relative to human PhD students [Schmidgall et al., 2025]; The AI Scientist (Sakana) hallucinated ablation tables; and of 200 Erdős solutions audited (from 700 attempted), only 13 (6.5%) were rated meaningfully correct. A 45-expert evaluation likewise positions current AI reviewers as complements to, not substitutes for, human experts—strong on aggregate criticism yet sharing blind spots and overlapping far more than independent humans do [Kim et al., 2026]. This unreliability is not confined to novelty scoring: LLMs asked to reverse-engineer black-box systems—a canonical scientific inference task—fail to do so reliably even with systematic query access [Geng et al., 2025]. The Leiden Declaration [Alper et al., 2026], signed by over 130 mathematicians, corroborates the indispensability of expert oversight: AI-produced proofs may contain almost-invisible errors that resist detection without human scrutiny, so no automated layer fully substitutes for V_{human} even where automated checking is most mature.

Proposed falsification test: The thesis is refuted if an automated validator matches a panel of ≥ 3 domain experts on novelty discrimination—Cohen's $\kappa \geq 0.80$ against the expert majority—on a benchmark of published-versus-withheld findings whose training-set membership is fixed by a pre-training cutoff date, across ≥ 3 distinct domains. Constructing the contamination-controlled benchmark and the expert baseline is itself an open task; a recent benchmark takes a step by restricting to post-2024 findings to control contamination [Liu et al., 2025], though it evaluates inspiration-based decomposition rather than closing the expert-baseline gap.

Novelty, significance, and correctness are distinct properties that current systems collapse into one scalar (e.g., The AI Scientist's 1–10 score [Lu et al., 2024]); a viable automated novelty V would need a contamination-controlled reference corpus, architectural separation between G and V, and a definition of "novel to the field" beyond lexical training-set membership (e.g. disruption or atypicality measures [Park et al., 2023b, Uzzi et al., 2013])—none of which current systems provide.

7.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no surveyed LLM-based system implements it: all deployed LLM-based policies are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty estimates over a tractable GNN posterior can be read as a coarse EIG surrogate—but only because DFT provides a well-calibrated computational V and the hypothesis space admits a learnable probabilistic model, conditions absent from the LLM-based systems here, where the "posterior" is implicit in model weights [Foster et al., 2021]. BED-LLM [Choudhury et al., 2026] derives a usable EIG posterior from an LLM's predictive distributions, but for conversational queries rather than experiment design—so the gap is integration with reliable E and V, not impossibility. Concretely, every LLM-based system in Table 1 degrades to heuristic π: Coscientist selects tools reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid. Recent falsification-driven approaches ground hypothesis selection in executable tests rather than free generation—AIGS selects hypotheses by their survival of automated refutation [Liu et al., 2024], and Popper runs sequential falsification experiments with statistical error control [Huang et al., 2025]—but neither computes or maximizes EIG: they substitute a falsification heuristic for the BED-optimal policy rather than realizing it.

Proposed falsification test: The gap is refuted if a system explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ whose hypothesis space admits no closed-form or finite-ensemble surrogate—a posterior defined directly over natural-language hypotheses or program text rather than a hand-specified low-dimensional model—thereby excluding tractable-model proxies such as GNoME's deep-ensemble policy.

Architecturally, with θ implicit in frozen weights and no per-experiment update, a system must either maintain an explicit probabilistic surrogate (GNoME's route, viable only where a calibrated input–outcome model exists) or use amortized EIG estimation [Foster et al., 2021], which remains unsolved for open-ended $\mathcal{S}$. Part of this gap is moreover conceptual: EIG-optimal selection presupposes an already-represented hypothesis space, whereas how genuinely new hypotheses or representations *arise* falls outside current BED formulations [Yanai and Lercher, 2019]—the generative counterpart to the non-identifiability that Proposition 3 establishes for novelty.

7.3 The Experiment Fidelity Gap

Test-time compute has created a structural asymmetry: hypothesis generation is now fast and cheap, while experiment execution E remains physically rate-limited—more reasoning cannot reduce DFT wall-time, wet-lab turnaround, or synthesis lags. Yet no surveyed system has a principled criterion for when to escalate from computational proxy E to physical experiment; result-caching accumulates past outcomes but does not determine *when* to invest in higher-fidelity measurement. Without an escalation criterion, computational loop closure does not imply physical discovery.

The pattern is consistent across domains. In materials science, GNoME predicts 380,000 stable compounds via DFT [Merchant et al., 2023], but only 736 have been independently experimentally realized [Merchant et al., 2023]: the escalation decision is made externally with no principled automation. In chemistry, ChemCrow and ChemReasoner close the computational loop but leave the wet-lab loop open: NMR and MS characterization require physical execution that no current system knows when to invoke [Gromski et al., 2019]. In biology, BioPlanner's V metrics are insufficient to close the loop autonomously [O'Donoghue et al., 2023]—the physical execution question is never reached. In cognitive science, Jagadish et al. [Jagadish et al., 2026] use a behavioral foundation model as proxy E (synthetic participant data in lieu of human subjects) but give no criterion for escalating to in-vivo validation, themselves flagging synthetic data as a hypothesis accelerator, not ground truth. GNoME is the most successful case precisely because it *fixes* E at a single fidelity (DFT) and optimizes within that constraint: it circumvents the escalation problem rather than solving it. ERA likewise optimizes within a fixed computational evaluator, generating expert-level empirical software judged by a held-out quality metric without physical escalation [Aygün et al., 2026].

This problem is structurally distinct from Op1 and Op2: a complete V evaluates novelty only at the fidelity it receives, and an EIG-optimal π selects among experiments at the currently available fidelity—so without E escalation criteria the system can close a computational loop indefinitely without reaching the physical evidence trustworthy scientific claims require.

Recent autonomous "self-driving" laboratories bear directly on this gap but do not close it. In materials science, the A-Lab [Szymanski et al., 2023] couples computational screening to robotic synthesis, initially reporting 41 of 58 target compounds synthesized, later revised after post-publication diffraction re-analysis to 36 of 40 reported successes confirmed (4 inconclusive) [Szymanski et al., 2026]—but it escalates by a fixed screen-then-synthesize rule rather than an evidence-informed criterion for when physical validation is warranted. Robotic chemistry platforms such as ORGANA [Darvish et al., 2024] automate the physical execution of wet-lab experiments, and self-correcting multi-agent systems such as AutoLabs [Panapitiya et al., 2025] automate the generation of hardware-ready protocols; reviews document the rapid maturation of the underlying hardware [Tobias and Wahab, 2025]. Yet automating E supplies capacity, not a criterion: these systems operate at a fixed or externally chosen fidelity and do not decide *when* accumulated computational evidence warrants escalating to a physical assay. The bottleneck is thus orthogonal to hardware automation—it persists even as the physical loop becomes mechanically executable.

Proposed falsification test: The gap is refuted if a system implements an evidence-informed criterion for escalating E from computational proxy to physical experiment and achieves a higher rate of experimentally validated discoveries per unit of physical experimental cost than a fixed-escalation system, evaluated on a pre-registered benchmark.

8 Discussion

Validator design as a constraint. Our analysis suggests V design appears to be a necessary constraint on *trustworthy* closed-loop operation: unreliable validation prevents a system from learning whether its outputs represent progress—an ecosystem-scale symptom is the documented surge in fabricated citations [Zhao et al., 2026], and strong task performance need not imply an internalized world model [Vafa et al., 2025]. Since the surveyed systems do not isolate V from G, E, M, and π, we read V completeness as a recurring bottleneck, not a causal priority—a hypothesis-generating framing, not an evidence-backed recommendation—and it suggests a diagnostic question: what level of V completeness does a domain support before optimizing G or π?

Human oversight. Rather than treating $H \in [0, 1]$ as a quantity to minimize, we read explicit human integration at the V_{human} and π levels as appropriate: human judgment accesses an information source no automated proxy yet replicates, particularly for the non-automatizable components of novelty and significance. The goal is not full autonomy but *appropriate* autonomy.

Towards collective AI co-scientists. All systems analyzed are single-agent loops; extending Servo to a multi-agent collective with shared memory w and a collective validator [Taniguchi et al., 2025] poses exactly the three open problems above as its core implementation challenges, with ARA [Liu et al., 2026] a plausible entry point for w. Such a collective must also preserve epistemic diversity: AI reviewers already overlap far more than humans [Kim et al., 2026], risking the transient diversity that benefits epistemic communities [Zollman, 2010].

Limitations. Servo abstracts away several factors: the cost structure of E (orders of magnitude across domains), the non-stationarity of S, and the single-agent framing's social layer [Taniguchi et al., 2025]. More fundamentally, the V-completeness pattern is a cross-system *association*, not an established causal determinant, and the surveyed systems are a convenience sample subject to selection and survivorship bias. Mechanical loop closure moreover does not require completeness (AI Scientist v1 and NovelSeek close a computational loop at low completeness), so the claim concerns *trustworthy* closure, which we do not measure independently of V. The coding methodology and multi-vendor reliability estimates are in Appendix B (calibration Fleiss' κ=0.79; holistic V-completeness ordinal only κ=0.39; no calibrated *automated* novelty gate). Using LLM coders to characterize systems whose validators we argue are unreliable is a reflexive limitation, which is why the central claim is pinned to the calibration sub-construct and human expert coding remains necessary future work.

9 Conclusion

We introduced Servo (S, G, E, V, M, π), connecting AI Scientist systems to POMDP and BED theory across six core systems and seven domains. Its propositions support only narrow claims, and the organizing hypothesis linking gating calibration to trustworthy closure remains provisional. Servo names the vocabulary for closing the remaining bottlenecks: validator completeness, search-policy design, and experiment-fidelity escalation.

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

Research Stage Assessment

1. **Hypothesis development.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: The core Servo framework tuple (S, G, E, V, M, π), the V-completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.

2. **Experimental design and implementation.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.

3. **Analysis of data and interpretation of results.**
   Answer: **[C]**
   Iteration Effort: **[Medium]**
   Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V-completeness thesis emerged from the human author's synthesis. Consistent with this venue's premise, the cross-system coding is AI-conducted under human direction and disclosed as such; its rigor rests on auditability (released rubric, per-row source quotes, and regeneration/citation-validation scripts) and conservative claim-scoping rather than an independent human gold standard, which we treat as orthogonal to the framework contribution and defer to future work.

4. **Writing.**
   Answer: **[C]**
   Iteration Effort: **[High]**
   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. **AI systems used.**
   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. **Observed AI Limitations.**
   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial

review rounds were needed to correct framing and remove overclaims. (3) Context drift: over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems). The propositions under explicit assumptions (§A) clarify the separation of observation, reward validation, and novelty judgment rather than establishing a V-completeness thesis, and the cross-domain survey is presented as illustration; the paper delimits both the assumptions and the survey's illustrative role in §A and §8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [Yes]

   Justification: Appendix A states three propositions, each with its assumptions (Assumption 1, Definition 1) and a proof sketch grounded in standard results on Bayesian experimental design and posterior consistency under model misspecification. The cross-domain survey is presented as illustration of these results, not as independent empirical proof.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiments. The cross-domain analysis approach (system selection and component extraction from primary sources) is described in §5 and §6; machine-readable extraction sheets (core systems and the seven-domain Table 2) with a representative source quote and citation per row and citation-validating regeneration scripts, explicit inclusion/exclusion criteria, multi-vendor blind coding over 14 systems with Fleiss' κ (plus a two-vendor Cohen's κ and recode sensitivity for the calibration re-coding), and a counterexample search are released as supplementary material (see §8). All analyzed papers are publicly accessible. The released coding should be treated as *provisional structured annotation*: the table-generation scripts verify citation presence and non-empty source quotes but do not validate semantic entailment of each coded field; independent reproduction of the interpretive coding requires consulting the cited primary sources directly.

5. **Open access to data and code**

   Question: Does the paper provide open access to data and code?

   Answer: [Yes]

   Justification: The reproducibility package is released as supplementary material in the `analysis/` directory: the coding protocol (`coding_protocol.md`); coding sheets with a representative source quote and citation per row for both the core systems (`systems.csv`) and the seven-domain Table 2 (`domain_systems.csv`); the multi-vendor calibration re-coding (`multicoder/`); the reliability report (`reliability_report.md`); and the citation-validating table-regeneration scripts (`build_servo_tables.py`, `build_domain_tables.py`, `compute_calib.py`). All analyzed papers are publicly accessible; no separate dataset was generated. The complete reproducibility package is additionally released at `https://github.com/ysys143/servo-caisc2026`.

6. **Experimental setting/details**

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or statistical significance information?

   Answer: [NA]

   Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [NA]

   Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. **Research Ethics**

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the experiment fidelity gap and novelty non-automatability).

A Formal Analysis of Validator Completeness

We make the role of V precise by separating two objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$, which drives the Bayesian belief update, and the *validator* $V(o)$, a reward estimate that the policy π uses to score outcomes. The results hold in a tractable subproblem; the open-ended regime is left open.

Assumption 1 (Tractable, identifying subproblem)**.** *(i) the search space S is bounded and the parameter space Θ is compact; (ii) T is stationary; (iii) the prior $p(\theta)$ has support containing the true $\theta^\star$; (iv) the observation model $p(o \mid \theta, d)$ is well-specified and identifies θ (for $\theta \neq \theta'$ some reachable design separates them); and (v) the realized design sequence is* sufficiently informative *for posterior consistency: infinitely often it selects a reachable design whose Kullback–Leibler separation between any two surviving hypotheses is bounded below by a fixed $\delta > 0$. We do not equate this with a divergent sum of adaptive EIG terms: incremental EIG is an expected entropy reduction bounded by initial entropy in finite-state settings, so cumulative EIG can be bounded even when the posterior concentrates. The condition here is on KL-separation between hypotheses, not on accumulated EIG.*

Definition 1 (Validator as reward channel)**.** *$V(o)$ estimates the true reward $R(s, a)$; it is* calibrated *if $\mathbb{E}[V(o) \mid s, a] = R(s, a)$ and* biased *if $\delta(s, a) = \mathbb{E}[V(o) \mid s, a] - R(s, a) \neq 0$ for some (s, a). V is distinct from the update likelihood $p(o \mid \theta, d)$ unless explicitly coupled into it.*

Proposition 1 (Sufficiency for closure)**.** *Under Assumption 1 the posterior concentrates on $\theta^\star$ (the loop closes), and the one-step EIG rule (Equation 1) is myopically optimal for the information objective—both independently of the validator, since V is a reward channel while EIG is a functional of the prior and likelihood alone.*

Proof sketch. A well-specified, identifying likelihood with *divergent* accumulated information on a compact Θ (Assumption 1(i),(iv),(v)) is the standard sufficient condition for Bayesian posterior consistency [Ghosal and van der Vaart, 2017], giving $b_t \to \delta_{\theta^\star}$. Calibration is required for neither consistency nor EIG-optimality (both depend only on the prior and likelihood); it concerns the separate reward channel (we make this precise below). The uniform-information clause (v) is essential: a sequence with $\mathrm{EIG}(d_t) > 0$ but $\sum_t \mathrm{EIG}(d_t) < \infty$ leaves the posterior short of a point mass even under identifiability and calibration. $\square$

Proposition 2 (Drift under misspecified coupling)**.** *Suppose the validator enters the belief update, i.e. the effective update is $q_\theta(o) \propto p(o \mid \theta, d)\ \exp(\beta V(o))$ with coupling $\beta > 0$. If V depends on the outcome in a θ-dependent (non-affine) way, the posterior concentrates on the Kullback–Leibler projection $\theta^\dagger$ [Berk, 1966], in general distinct from $\theta^\star$: the loop closes computationally but potentially on a different target. A θ-independent additive constant in V cancels in the log-partition and leaves $\theta^\star$ unchanged (a scale change instead rescales the coupling β); and when V is a pure policy channel ($\beta = 0$) no validator transformation moves the target—it can at most degrade design choice and slow information acquisition (interacting with Assumption 1(v)).*

Proof sketch. A coupled, misspecified update likelihood places the posterior at its KL minimizer [Berk, 1966]; the log-partition $\log \int p(o \mid \theta, d)\, e^{\beta V(o)}\, do$ relocates that minimizer iff its dependence on θ changes, which a θ-independent additive constant does not. With $\beta = 0$ the update likelihood is unchanged, so $\theta^\star$ is invariant to V. $\square$

Proposition 3 (Non-identifiability of novelty)**.** *Let θ_{nov} be a novelty/significance component that is a priori independent of the identified components. If the channel likelihood of every automated validator is independent of θ_{nov}, then no automated validator admits a consistent estimator of θ_{nov}, and loop closure on novelty requires an additional channel informative about θ_{nov}, currently supplied only by V_{human}.*

Proof sketch. If the likelihood does not depend on θ_{nov} and the prior does not couple it to identified components, the posterior marginal on θ_{nov} equals the prior for all data; θ_{nov} is unidentified and admits no consistent estimator. $\square$

The formal results support a narrower distinction: posterior closure in the tractable subproblem depends on a well-specified, identifying, sufficiently informative observation model (Proposition 1), while validator calibration governs the reward or acceptance signal used by the policy—not posterior consistency, and not which design maximizes EIG (information-optimal and reward-optimal designs coincide only when validator reward preserves the EIG ordering, and can diverge even for a perfectly calibrated V). Proposition 2 shows that drift requires an incorrectly *coupled* validator, not a mere ranking shift; Proposition 3 isolates the one component no automated validator can resolve. These results do not by themselves prove that the empirical V-completeness ordinal is necessary for all trustworthy autonomous discovery; that claim remains a design hypothesis requiring independently coded outcomes. The surveyed systems should therefore be read as motivating examples for separating observation, reward validation, and novelty judgment—not as a formal demonstration that V-completeness is the primary determinant of closure.

B Cross-System Coding Reliability

The cross-system survey is released as provisional structured annotation. The claim concerns *trustworthy* closure, which we do not measure independently of V: the `trustworthy_closure` coding draws on outcome signals only partly outside V (peer review and publication overlap V_{human}) and, at six systems with outcomes entangled with V, cannot test the direction. For traceability we release the rubric, coding sheets with a representative source quote and citation per row (core systems and the cross-domain Table 2), and validator scripts that reject any uncited or unquoted row; the scripts verify formal presence of quotes and citation keys but not semantic entailment, so per-field judgment still requires the cited primary source.

Three independent, model-pinned, blind LLM-based coder agents from different vendors coded 14 Tier-1 systems (see `multicoder/target_systems.md`): Claude Code (`claude-opus-4-8`, Anthropic), Codex CLI (`gpt-5.5`, OpenAI), and Antigravity (`gemini-3.1-pro`, Google), each run headless and model-pinned, with exact invocations and per-model outputs released in the supplement. The six core systems in Table 1 are a representative analytical sample spanning the discovery-loop spectrum. Inter-coder agreement (Fleiss' κ over the 13 systems coded by all three vendors—NovelSeek drew only two usable codings after a coder timeout and is excluded) is substantial-to-perfect on most constructs (calibration κ=0.79, loop status κ=0.74, semantic-validation presence κ=1.0; mean κ=0.68) and only fair on the holistic V-completeness ordinal (κ=0.39), on which Cohen's κ against the author's reference labels is negative (-0.25 to -0.43)—the one construct the central claim does not rest on (it rests on the calibration sub-construct, κ=0.79). All reported as automated transparency, not human-level reliability. The low ordinal agreement plausibly reflects that a single ordinal compresses distinct properties—gate calibration, property coverage, experimental fidelity, and external independence—that a vector-valued treatment would separate; we therefore read the V-completeness ordinal as exploratory rather than a central measure.

A counterexample search found no V-incomplete system achieving *trustworthy* closed-loop discovery (the lone near-miss, AI Scientist v1, closes on a biased validator but yields unreliable output). A two-vendor recode over all 14 systems (supplement) resolves a *present*-vs-*gating* ambiguity in the calibration construct—the reference labels and both coders agree the layer is present but non-gating (raw agreement 13/14, Cohen's κ=0.86; a robustness check, not expert-validated reliability)—and finds no system with a calibrated *automated* novelty gate, grounding the novelty open problem. Introduced post-hoc, this $V_{\text{present}}/V_{\text{gating}}$ split reflects iterative construct development rather than a prospective reliability study.

C Per-Domain Analysis

This appendix expands Table 2 into a per-domain reading in the Servo vocabulary, stating for each field the validator type and its reliability, the loop status that validator permits, the search policy actually deployed, and the bottleneck that follows. The coding is released as the machine-readable `domain_systems.csv`, with a representative source quote and citation keys per row; the entries below paraphrase those quotes and should be read as a structured reading of the cited work rather than an independent empirical study.

Mathematics, formal (V_{formal}; fully closed). FunSearch [Romera-Paredes et al., 2024] and DeepSeek-Prover [Xin et al., 2024] validate against a formal or numeric oracle—an evaluator function or proof kernel returning a deterministic pass/fail. This V_{formal} provides deterministic formal-correctness feedback, enabling mechanically closed evaluation within the formalized task; the loop is fully closed. The residual bottleneck is not validation but S: pretraining contamination makes it hard to certify that a discovered result is genuinely novel rather than memorized—the formal-domain instance of the novelty problem (Proposition 3).

Mathematics, natural language ($V_{\text{sem}} + V_h$; closed at low quality). Aletheia [Feng et al., 2026] produces natural-language proofs scored by LLM and human judgment. The loop closes mechanically, but V_{sem} is imperfect: of 200 audited Erdős solutions only 13 (6.5%) were rated meaningfully correct. Replacing the formal oracle with a semantic one reintroduces V_h and exposes hallucination and plagiarism as failure modes—an illustration of how a miscalibrated reward channel admits drift (Proposition 2). The Leiden Declaration [Alper et al., 2026] corroborates this from the practitioner side: AI-produced natural-language proofs may contain almost-invisible errors that resist detection without expert scrutiny, reinforcing that V_{human} is not replaceable even where automated checking exists.

Physics ($V_{\text{emp}} + V_h$; none/partial). AI Feynman [Udrescu and Tegmark, 2020] fits equations to data with deterministic search, while symbolic-model discovery [Cranmer et al., 2020] distills equations from trained neural networks via stochastic symbolic regression (genetic algorithms). Validation is empirical and of medium reliability, but the systems carry no memory transfer across problems: each task starts fresh. The bottleneck is thus the absence of cross-problem memory transfer: each task begins from scratch without accumulated experimental patterns from prior discoveries.

Chemistry (V_{emp} surrogate; computational loop only). ChemReasoner [Sprueill et al., 2024] validates computationally with surrogate signals and searches by beam, whereas ChemCrow [Bran et al., 2024] plans via ReAct tool-use and validates in part through physical wet-lab execution on a robotic platform (with expert and LLM evaluation). The computational loop closes, but physical characterization (NMR, MS; [Kamber, 2026]) leaves the wet-lab loop open: the validator that would confirm a real discovery sits outside the automated boundary, so closure is partial at the measurement interface.

Materials science ($V_{\mathbf{emp}}$ via DFT; computationally closed). GNoME [Merchant et al., 2023] runs DFT-in-the-loop active learning, predicting 2.2 million structures stable with respect to prior work (380,000 on the updated convex hull), with uncertainty estimates that the manuscript reads as a coarse approximation to the BED ideal within the computational surrogate. DFT should be read as an approximation of physical energies rather than a fully calibrated empirical oracle for materials discovery: of the predicted stable structures, only 736 have been independently experimentally realized [Merchant et al., 2023], and critics note that predicted compounds should not be equated with validated materials exhibiting demonstrated functionality [Cheetham and Seshadri, 2024]. Servo therefore codes this domain as computationally closed for phase-stability screening, with the physical materials-discovery loop only partly closed. The remaining bottlenecks are resource-bound: DFT wall-time and the lag before predicted structures are synthesized and characterized.

Biology ($V_{\mathbf{emp}} + V_h$; partial). BioPlanner [O'Donoghue et al., 2023] generates experimental protocols, but the validation signals are weak and biological-validity metrics are insufficient to close the loop autonomously, so reliability is low-to-medium with effectively no search policy beyond generation. The bottleneck is V itself: without metrics that track ground-truth biological validity, the loop cannot be closed on anything trustworthy.

Social science (statistical tests; partial). Automated-social-science pipelines [Manning et al., 2024] validate with statistical tests over full-factorial designs, while LLM-simulation studies [Aher et al., 2023, Park et al., 2023a] validate simulated behavior with statistical tests over ablation and controlled comparisons. The characteristic failure is circular validity: when the generator G and the simulated subjects share a model, the validator measures the model's self-consistency rather than an external fact—a violation of the identifying-channel condition of Assumption 1.

Computer science / algorithms ($V_{\mathbf{emp}}$; fully closed). AutoML-Zero [Real et al., 2020], algorithm discovery [Surina et al., 2025], and AutoKernel [Jaber and Jaber, 2026] close the loop via reliable benchmark/empirical validation and evolutionary or LLM-guided search. Validation is high-reliability and the loop is fully closed; the residual bottleneck is interpretability—the discovered artifacts work but resist human understanding, a limitation of M and output form rather than of V.

Cross-domain reading. The bottleneck each field hits is largely a function of its V type: a perfect formal oracle yields a fully closed, contamination-limited loop; an imperfect semantic or empirical V yields hallucination, circular validity, or insufficient-metric failures; and a reliable but expensive empirical oracle (materials) shifts the bottleneck from epistemics to resources. This is the diagnostic use of Table 2: it does not claim a tested cross-domain law, but shows that the same (S, G, E, V, M, π) vocabulary localizes where each domain's loop breaks. V type shapes whether a domain's loop can close *trustworthily*; the search policy π independently shapes what is *scientifically discoverable*—whether the loop reaches genuinely unexplored regions of S or remains within the already-exploited frontier. Table 2 captures the V dimension of this two-axis picture; the π dimension remains uniformly underdeveloped across all surveyed domains, with materials science the sole partial exception.